Simulation Exercises

Deadline: 27 September 2024

Exercise 1: Simulation of a Falling Droplet

Simulate the behavior of a water droplet with a diameter of 1 mm falling from a height of 3.5 mm. The following assumptions and conditions apply:

- Incompressible, laminar flow conditions.
- Physical properties at normal atmospheric temperature (298 K):
  - Density of water: 998.2 kg/m^3.
  - Density of air: 1.225 kg/m^3.
- Dynamic viscosity:
  - Water: 0.001 Pa·s.
  - Air: 0.0000181 Pa·s.
- Gravitational acceleration: 9.81 m/s^2.

Your task:

1. Simulate the falling droplet using the 'interFoam' solver in OpenFOAM.
2. Identify an appropriate test case from the tutorial folder.
3. Axisymmetric boundary conditions may be used for computational efficiency.

4. Generate the mesh using the ‘blockMeshDict‘ utility.
5. Set the initial condition for the droplet using the ‘setFields‘ utility to modify the alpha value.
6. Perform the simulation with the correct solver.
7. Plot the trajectory of the droplet’s centroid over time using ParaView.

Exercise 2: Simulation of a Rising Bubble

Simulate the behavior of an air bubble with a radius of 0.25 mm rising in water. The assumptions and conditions remain the same as in Exercise 1:

- Incompressible, laminar flow conditions.
- Densities and viscosities of water and air as specified in Exercise 1.
- Gravitational acceleration: 9.81 m/s^2.

Your task:

1. Simulate the rising bubble using the ‘interFoam‘ solver.
2. Select an appropriate test case from the tutorial folder and adapt it for this scenario.
3. Axisymmetric boundary conditions may be considered.
4. Generate the computational mesh using the ‘blockMeshDict‘ utility.
5. Set the initial conditions for the bubble using the ‘setFields‘ utility.
6. Run the simulation with the appropriate solver.
7. Plot the bubble’s centroid position over time using ParaView.

ParaView Task

In both exercises, you are required to create a Python script in ParaView that extracts the centroid position of the droplet or bubble over time. The script should output a CSV file with the following columns:

- Time (s)
- Centroid position (mm)

Physical Model Assumptions

- Incompressible, laminar flow.
- Gravitational acceleration: 9.81 m/s^2.
- Normal temperature: 298 K (25°C).
- Density of water: 998.2 kg/m^3.
- Density of air: 1.225 kg/m^3.
- Dynamic viscosity of water: 0.001 Pa·s.
- Dynamic viscosity of air: 0.0000181 Pa·s.

Submission Instructions

1. Please submit a single ZIP file containing all the necessary OpenFOAM files and Python scripts.
2. Additionally, submit a single PDF file containing:
   - Simulation results and plots.
   - A brief description of your approach and findings.
3. Both files (ZIP and PDF) should be submitted to the email: iterasim@gmail.com.